Step 4 — Clinical Feasibility Report

TRUSTXAI · Spandan Kumar Gope · 22MIP10039

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Status: Measured feasibility gates passed; application and thesis experiments not yet implemented.

What was actually tested

The confirmed application is a diagnostic Clinical Decision Support System research prototype with a ChatGPT/Codex-inspired interface. The benchmark task selects the most likely diagnosis from four supplied options. An exam-vignette benchmark is not real-patient clinical validation. The user confirmed no API access/budget; no GPT/Gemini outputs or comparison results exist.

The primary diagnosis-only cohort was refined by question semantics before its predictions. Two exploratory selections are archived. The final 24-case feasibility pool excludes all 48 earlier source IDs. The old broader-cohort 16/24 score and the primary 16/24 score belong to different samples and must not be pooled or treated as replicated final-test performance.

The final diagnosis-only cohort has 917 candidate training and 117 candidate test cases before grouping/eligibility. After archived exploratory sources and declared budgets are excluded, 877 training and 117 test cases remain. The final grouping found 0 exact repeats and 0 near-duplicate edges, with 0 cross-partition cases quarantined. The 24/40/60 manifest has no overlapping source components. Maximum selected task prompt: 552 tokens; cap: 1,536. Manifest SHA256: 730d9ecfa80b04f5bc466f74d9750a5be08c35c029dc32c63ee21d11f4e70bff. All these operations preceded primary feasibility predictions.

Observed gates

Check	Required	Actual observation
Diagnostic-option accuracy	At least 0.50 on 24 sources	16/24 = 0.667
Position macro-F1	At least 0.40; all positions nonzero recall	0.6702; all positive
Valid answer letters	At least 95%	100%
Median target response	At most 20 seconds	12.12 seconds
Joint JSON format	At least 80% on 8 requested sources	8/8 = 1.000
Median pair generation	At most 180 seconds	19.77 seconds
Peak inference RSS	At most 24 GiB	18.91 GiB across separately loaded phases
Projected fixed workload	At most 24 aggregate CPU-hours	21.10 CPU-hours with headroom

The selected fixed target is **Qwen/Qwen3-4B-Instruct-2507**, revision cdbee75f17c01a7cc42f958dc650907174af0554, CPU FP32 with four threads. Its primary diagnosis-only feasibility pool has 24 fresh cases, six per correct option position, disjoint from all 48 archived exploratory selections. It returned **16/24 correct (66.7%)**, option-position macro-F1 **0.6702**, and 24/24 valid letters. Median response: **12.12 seconds**; peak RSS: **18.91 GiB**. Position recalls A/B/C/D are 0.833, 0.667, 0.500, 0.667. This is operational feasibility, not final-test accuracy or clinical validation.

The communication model is **Qwen/Qwen3-1.7B-GGUF Q8_0**, official publisher revision 90862c4b9d2787eaed51d12237eafdfe7c5f6077, using llama-cpp-python 0.3.16 on CPU with two decode and four prefill threads. Exact published artifact SHA256 is recorded; its original conversion commit is not established. The reference tokenizer was checked for exact native token IDs, and each actual generation also checks them. Models and roles are separated; communication does not change the frozen target decision.

On the fixed eight feasibility cases, 16 preselected omissions were evaluated; 3 changed the model answer and 8/8 selected-result rechecks agreed. Informative evidence was available for 8/8 cases. Joint JSON was valid for **8/8** requested pairs; median generation was **19.77 seconds**, with native-phase peak RSS **2.39 GiB**. These measurements cover runtime and format. They do not establish that every generated sentence preserves its evidence. No repair, independent semantic audit, H1/H2/H3 comparison or final-test inference has run.

Hardware and failed attempts

The cloud restarted during this work. The current CPU has AVX2 without AVX512-BF16; preliminary BF16 clinical queries took 65–77 seconds and that incomplete profile was stopped. The selected target uses FP32. A native Q8 communication model was built from pinned public source for CPU compatibility; no TLS, source hash or model-integrity check was disabled. Target and communication run in separate phases, rather than claiming a measured simultaneous-memory figure. The user's MacBook Air M5 was not benchmarked.

The original diagnosis-keyword filter admitted workup, mechanism and treatment questions. Manual question review corrected this with positive diagnostic-request grammar, and laboratory tables after a question were moved into case evidence. These are disclosed scope/input corrections, not exclusions based on final-test outcomes. Source text, exploratory outputs and manifests are preserved outside the active protocol. The final cohort/prompt hash was frozen before its primary predictions.

Budget and next boundary

The fixed workload estimates evidence for 40 development plus 60 test cases, at most 400 unique joint narrative generations after exact shared-initial reuse, 100 natural-output critic calls, 1,200 diagnostic pairs and up to 160 natural P/development check pairs. Exact NLI-input keys were counted on 96 unreviewed cost-layout variants derived from the eight feasibility records; they are not semantically labelled diagnostic results. No cross-source cache benefit is assumed. Original-query mean cost is used as a floor for omitted/rechecked queries.

The measured-cost projection is **21.10 aggregate process CPU-hours**, including 25% headroom; estimated elapsed time is **7.32 hours**. The 24-CPU-hour expected-cost gate passed: **True**. This is a projection, not an executed matrix. An uncached full-output/full-context stress scenario is **77.86 CPU-hours**, so the budget is not a worst-case guarantee. A changed diagnostic layout requires a fresh projection. Checkpoint before the ceiling, preserve pending cases and document any allocation change; do not drop hard cases. Extra exploratory tuning and human annotation are outside this fixed-workload estimate and must be logged separately.

All observed feasibility gates passed: **True**. Neither the JSON parser nor the native tokenizer probe establishes semantic explanation correctness. There are zero development/test target predictions, zero completed six-condition thesis comparisons and zero recruited independent annotators. Medical effectiveness, calibrated confidence, patient benefit and superiority over GPT/Gemini have not been established.

The strongest planned comparison uses the same target, communicator and shared initial pair for grounded blind rewrite versus TRUSTXAI checking/repair. Independent evidence-based labels are still needed. Templates are a strong semantic-output comparator; their LLM-generation yield is not applicable, and a template fallback is not counted as successful generated narrative.

Step 5 implements the actual application, full all-span evidence pipeline, checkers, repair and baseline adapters. The design brief specifies a sidebar, conversation, rounded composer, light/dark themes and an evidence drawer. Application code, a final research paper and Review 2 slides are not represented as completed by this report.